ANALYTICAL CUMULATIVE EXAM
March 4, 1995

Acid-Base Chemistry

(30 pts) 1. You are given the dissociation equilibrium of a monoprotic weak acid HA in water (Equation 1),

$$HA + H_2O = H_3O^+ + A^- \qquad (1)$$

(a) Derive the following form of the Henderson-Hasselbalch equation (Equation 2), where K_a is the acid dissociation constant.

$$pH = pK_a + \log \frac{[A^-]}{[HA]} \qquad (2)$$

(b) The dissociation of water itself, Equation 3, can play a role in acid/base reactions and equilibria.

$$2\,H_2O = H_3O^+ + OH^- \qquad (3)$$

What assumptions about the dissociation of water do you have to make so that Equation 2 is valid? (Note that Equation 3 was not considered in the derivation of Equation 2.)

(c) Rewrite Equation 2 in terms of K_b and pOH instead of K_a and pH. (Note that the K_b is the base dissociation constant for $A^- + H_2O = OH^- + HA$. Also note that $K_w = K_a K_b$, where $K_w = 10^{-14}$.)

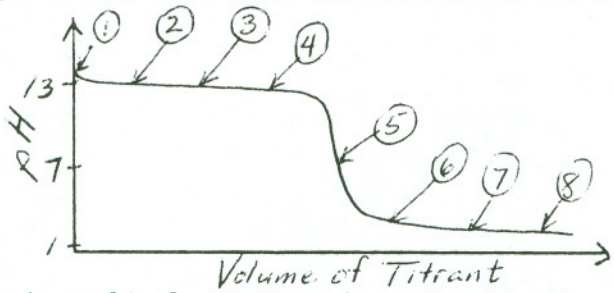
(40 pts.) 2. The titration of a solution of a weak monobasic base, A^-, with a strong acid is monitored with a pH electrode and meter. The resulting titration curve is shown below.

(a) Identify the points at which $pH = pK_a$ and at which $pOH = pK_b$. Explain your reasoning.

(b) Identify the endpoint of the titration. Explain your reasoning.

(c) Identify the buffer region. Explain your reasoning.

(d) Predict what changes would occur in the titration curve if the starting weak base solution were ten times more dilute?



(15 pts.) 3. An indicator dye is used in the titration of a weak acid with a strong base. The indicator dye is a weak acid (HIn), itself, but there is a striking difference in the color between the acid (HIn) and conjugate base (In^-) forms.

$$\underset{\text{colorless}}{HIn} + H_2O = H_3O^+ + \underset{\text{red}}{In^-}$$

The K_a of the indicator is **7.94×10^{-10}**. The eye can see a pure color (either red or colorless) when there is a **10 to 1 or a 1 to 10 ratio** of the two forms of the indicator. What color would you expect to see at **pH of 7.8**? Show your work.

(15 pts.) 4. Coulometry can be used to replace a buret in an acid/base titration. Below are possible cathodic and anodic reactions that can occur at an electrode in contact with an aqueous solution during coulometry.

$$2\,H^+ + 2\,e^- = H_2 \text{ (cathode)}; \quad H_2O + 1\,e^- = OH^- + 1/2\,H_2 \text{ (cathode)}$$

$$H_2O = 2\,e^- + 1/2\,O_2 + 2\,H^+ \text{ (anode)}; \quad 2\,OH^- = 2\,e^- + 1/2\,O_2 + H_2O \text{ (anode)}$$

Quantitatively describe a way in which you could make *at least* **10 mL** of a NH_3/NH_4^+ buffer of **pH = 9.07**. Note that you only have access to **0.300 M NH_4Cl**, a constant current coulometer, and any electrodes or containers for performing coulometry. K_b for NH_3 is **1.76×10^{-5}**. Your coulometer can be set to a constant current anywhere between **1 μA and 1 A**. Note that Faraday's constant, F, is **96485 Coul of e^-/mol e^-**.